

RYANN M. PEREZ

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SUMMARY

Computational biologist with 5+ years applying machine learning, generative AI, and molecular simulations to drug discovery and protein engineering. Co-author on 10 publications, including first-author work pioneering protein language models for amyloidogenesis and educational AI. NIH F31 fellow seeking to build novel, high-impact AI systems in biotech.

EDUCATION

PhD, Computational Biology

April 2026

University of Pennsylvania • GPA: 3.99 / 4.00

B.S. Chemistry, Minor in Biochemistry

May 2020

University of Delaware • GPA: 3.72 / 4.00

RESEARCH EXPERIENCE

University of Pennsylvania

September 2020 – April 2026

Graduate Researcher • Lab of Dr. E. James Petersson

- Developed custom protein language models achieving state-of-the-art performance on biologic aggregation and protein stability prediction; scaled inference pipeline to 64 million datapoints.
- Built and deployed an open-source LLM-based Q&A system for graduate-level chemistry; demonstrably improved learning outcomes and adopted by faculty for ongoing classroom use.
- Authored Python packages for experimental analysis, drug discovery, and protein simulations, contributing to 10 coauthored publications in 5 years.

SELECTED AWARDS & PRESENTATIONS

- NIH Predoctoral Individual NRSA (F31)** — Awarded by the National Institutes of Health for an original proposal on amyloid polymorphism determination (Sep 2024).
- UPenn Dean's Scholar** — One of 20 students in the School of Arts and Sciences recognized for outstanding academic achievement and intellectual promise (Apr 2024).
- Lectures on Generative AI in Chemistry** — Designed and led seminars for 50+ professionals and scientists on integrating ChatGPT into chemistry workflows (May 2025).

TECHNICAL SKILLS

Programming & ML: Deep learning, CNNs, Transformers, TensorFlow, PyTorch, Python, Classical ML, scikit-learn, machine learning models, Fine-tuning, LLMs, classification, NumPy, Pandas, Statistical analysis, HuggingFace, Agentic coding (Claude Code)

Biology: High-throughput assay design, Biomarker discovery, Bioinformatics, Genomics, Fluorescence polarization, Enzymatic assays, Protein expression & purification, FPLC, SDS-PAGE, MALDI-TOF MS

Computational Chemistry: ML for protein stability/solubility/aggregation, ADMET prediction, Custom virtual ligand screening, Structure-activity relationships, PyMOL, Rosetta / PyRosetta, PyOpenMS

Chemistry: LC-MS, HPLC, NMR, Peptide synthesis, Organic synthesis, Flash chromatography

SELECTED PUBLICATIONS

- Perez, R. M.**; Shimogawa, M.; *et al.* Large Language Models for Education: TAsk — An Open-Source Paradigm for Automated Q&A in the Graduate Classroom. *Comput. Educ.: Artif. Intel.* **2026**. DOI: 10.1016/j.caeai.2026.100546
- Li, X.; **Perez, R. M.**; Mach, R. H.; Giannakoulis, S.; Petersson, E. J. Machine Learning Prediction of Multiple Distinct High-Affinity Chemotypes for α -Synuclein Fibrils. *Chem. Commun.* **2026**. DOI: 10.1039/D5CC06228D
- Li, X.; **Perez, R. M.**; *et al.* Accurate Prediction of Protein Tertiary and Quaternary Stability Using Fine-Tuned Protein Language Models and Free Energy Perturbation. *Int. J. Mol. Sci.* **2025**. DOI: 10.3390/ijms26157125
- Perez, R. M.**; Li, X.; Petersson, E. J.; Giannakoulis, S. AggBERT: Best-in-Class Prediction of Hexapeptide Amyloidogenesis with a Semi-Supervised ProtBERT Model. *J. Chem. Inf. Model.* **2023**. DOI: 10.1021/acs.jcim.3c00817

Full publication list: scholar.google.com/citations?user=nOgqduAAAAJ